

Scenario 1 output 2 ollama

Here is the detailed threat modelling analysis for the LLM-Integrated Authentication Assistant System:

****STRIDE-based Threats****

1. ****Threat ID:**** STR01

****Threat Name:**** Unauthorized Access

****STRIDE Category:**** T (Tampering)

****Affected Component/Data Flow:**** P2, User Database, Session Storage

****Attack Scenario:**** An attacker submits a malicious login request to bypass authentication.

****Security Impact:**** Compromise of user credentials and potential access to sensitive data.

****Severity:**** High

****Likelihood:**** Medium

****Mitigation Strategy:**** Implement robust authentication mechanisms in P2 and ensure secure communication with the User Database and Session Storage.

2. ****Threat ID:**** STR02

****Threat Name:**** Data Tampering

****STRIDE Category:**** T (Tampering)

****Affected Component/Data Flow:**** P3, Session Management Module, Session Storage

****Attack Scenario:**** An attacker manipulates session data to gain unauthorized access.

****Security Impact:**** Compromise of session security and potential access to sensitive data.

****Severity:**** High

****Likelihood:**** Medium

****Mitigation Strategy:**** Implement secure session management in P3 and ensure proper validation of session data in Session Storage.

****LLM-specific Security Threats****

1. ****Threat ID:**** LLM01

****Threat Name:**** Malicious LLM Output

****Affected Component/Data Flow:**** LLM Integration Service, External LLM Provider

****Attack Scenario:**** The external LLM provider generates malicious output that is not caught by the application-level response handling.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.

****Severity:**** High

****Likelihood:**** Low

****Mitigation Strategy:**** Implement proper validation and sanitization of LLM output in the LLM Integration Service.

2. ****Threat ID:**** LLM02

****Threat Name:** Prompt Injection**
****Affected Component/Data Flow:** P5, LLM Integration Service, External LLM Provider**
****Attack Scenario:** An attacker injects malicious prompts to manipulate the external LLM's behavior.**
****Security Impact:** Compromise of system integrity and potential unauthorized access.**
****Severity:** High**
****Likelihood:** Medium**
****Mitigation Strategy:** Implement robust input validation in P5 and ensure proper sanitization of prompts before sending them to the External LLM Provider.**

3. ****Threat ID:** LLM03**

****Threat Name:** AI Component Evasion**
****Affected Component/Data Flow:** AI Chatbot Component, User Database, Session Storage**
****Attack Scenario:** An attacker exploits a vulnerability in the AI component to evade authentication and access sensitive data.**
****Security Impact:** Compromise of system integrity and potential unauthorized access.**
****Severity:** High**
****Likelihood:** Low**
****Mitigation Strategy:** Implement robust security controls for the AI component and ensure proper validation of user inputs.**

****OWASP LLM Top 10 Relevant Threats****

1. ****Threat ID:** OWASP01**

****Threat Name:** Insecure Input Validation**
****Affected Component/Data Flow:** P1, Web Frontend Interface**
****Attack Scenario:** An attacker submits malicious input that is not caught by the application-level validation.**
****Security Impact:** Compromise of system integrity and potential unauthorized access.**
****Severity:** High**
****Likelihood:** Medium**
****Mitigation Strategy:** Implement robust input validation in P1 to prevent malicious input.**

2. ****Threat ID:** OWASP02**

****Threat Name:** Sensitive Information Disclosure**
****Affected Component/Data Flow:** P2, User Database**
****Attack Scenario:** An attacker gains unauthorized access to sensitive user information.**
****Security Impact:** Compromise of user data and potential identity theft.**
****Severity:** Critical**
****Likelihood:** Low**
****Mitigation Strategy:** Implement robust access controls in P2 and ensure proper encryption of sensitive user data.**

3. ****Threat ID:** OWASP03**

****Threat Name:** Insecure Output Handling**

****Affected Component/Data Flow:**** LLM Integration Service, External LLM Provider
****Attack Scenario:**** The external LLM provider generates output that is not properly validated or sanitized by the application.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement proper validation and sanitization of LLM output in the LLM Integration Service.

4. ****Threat ID:**** OWASP04

****Threat Name:**** Lack of Secure Default Settings
****Affected Component/Data Flow:**** AI Chatbot Component, User Database
****Attack Scenario:**** The AI component is configured with insecure default settings that compromise system integrity.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement secure default settings for the AI component.

5. ****Threat ID:**** OWASP05

****Threat Name:**** Inadequate Data Validation
****Affected Component/Data Flow:**** P3, Session Management Module, Session Storage
****Attack Scenario:**** An attacker manipulates session data to gain unauthorized access.
****Security Impact:**** Compromise of session security and potential access to sensitive data.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust data validation in P3 to prevent malicious input.

6. ****Threat ID:**** OWASP06

****Threat Name:**** Excessive Data Collection
****Affected Component/Data Flow:**** AI Chatbot Component, User Database
****Attack Scenario:**** The AI component collects excessive user data that compromises system integrity.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement secure data collection policies for the AI component.

7. ****Threat ID:**** OWASP07

****Threat Name:**** Inadequate Error Handling
****Affected Component/Data Flow:**** LLM Integration Service, External LLM Provider
****Attack Scenario:**** The external LLM provider generates output that is

not properly handled by the application.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.

****Severity:**** High

****Likelihood:**** Low

****Mitigation Strategy:**** Implement robust error handling in the LLM Integration Service.

8. ****Threat ID:**** OWASP08

****Threat Name:**** Insufficient Logging and Monitoring

****Affected Component/Data Flow:**** AI Chatbot Component, User Database, Session Storage

****Attack Scenario:**** The system lacks sufficient logging and monitoring capabilities to detect potential security threats.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.

****Severity:**** High

****Likelihood:**** Low

****Mitigation Strategy:**** Implement robust logging and monitoring capabilities for the AI component.

9. ****Threat ID:**** OWASP09

****Threat Name:**** Inadequate Secure Coding Practices

****Affected Component/Data Flow:**** P1, Web Frontend Interface

****Attack Scenario:**** The application lacks secure coding practices that compromise system integrity.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.

****Severity:**** High

****Likelihood:**** Medium

****Mitigation Strategy:**** Implement robust secure coding practices for the application.

10. ****Threat ID:**** OWASP10

****Threat Name:**** Inadequate Secure Configuration Management

****Affected Component/Data Flow:**** AI Chatbot Component, User Database

****Attack Scenario:**** The system lacks secure configuration management capabilities that compromise system integrity.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.

****Severity:**** High

****Likelihood:**** Low

****Mitigation Strategy:**** Implement robust secure configuration management policies for the AI component.

****Threats related to Trust Boundaries****

1. ****Threat ID:**** TB01

****Threat Name:**** Data Exfiltration through External LLM Provider

****Affected Component/Data Flow:**** External LLM Provider, User Database

****Attack Scenario:**** The external LLM provider exfiltrates sensitive data from the system.

****Security Impact:**** Compromise of user data and potential identity theft.
****Severity:**** Critical
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the External LLM Provider.

2. ****Threat ID:**** TB02

****Threat Name:**** Unauthorized Access to AI Component
****Affected Component/Data Flow:**** AI Chatbot Component, User Database
****Attack Scenario:**** An attacker gains unauthorized access to the AI component.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust security controls for the AI component.

****Threats related to Prompt Injection****

1. ****Threat ID:**** PI01

****Threat Name:**** Malicious Prompt Injection through Web Frontend Interface
****Affected Component/Data Flow:**** P1, LLM Integration Service
****Attack Scenario:**** An attacker injects malicious prompts through the web frontend interface.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust input validation in P1 to prevent malicious prompt injection.

2. ****Threat ID:**** PI02

****Threat Name:**** Malicious Prompt Injection through AI Chatbot Component
****Affected Component/Data Flow:**** AI Chatbot Component, LLM Integration Service
****Attack Scenario:**** The AI component is exploited to inject malicious prompts into the system.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the AI component.

****Threats related to Sensitive Information Disclosure****

1. ****Threat ID:**** SID01

****Threat Name:**** Unauthorized Access to User Database
****Affected Component/Data Flow:**** P2, User Database
****Attack Scenario:**** An attacker gains unauthorized access to the user database.

****Security Impact:**** Compromise of user data and potential identity theft.
****Severity:**** Critical
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the User Database.

2. ****Threat ID:**** SID02

****Threat Name:**** Unauthorized Access to Session Storage
****Affected Component/Data Flow:**** P3, Session Storage
****Attack Scenario:**** An attacker gains unauthorized access to the session storage.
****Security Impact:**** Compromise of session security and potential access to sensitive data.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust security controls for the Session Storage.

****Threats related to Insecure Output Handling****

1. ****Threat ID:**** IOH01

****Threat Name:**** Malicious LLM Output through External LLM Provider
****Affected Component/Data Flow:**** External LLM Provider, LLM Integration Service
****Attack Scenario:**** The external LLM provider generates malicious output that is not properly handled by the application.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the External LLM Provider.

2. ****Threat ID:**** IOH02

****Threat Name:**** Malicious LLM Output through AI Chatbot Component
****Affected Component/Data Flow:**** AI Chatbot Component, LLM Integration Service
****Attack Scenario:**** The AI component is exploited to generate malicious output into the system.
****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the AI component.

****Threats related to Excessive Agency****

1. ****Threat ID:**** EA01

****Threat Name:**** Overprivileged AI Component
****Affected Component/Data Flow:**** AI Chatbot Component, User Database
****Attack Scenario:**** The AI component has excessive privileges that compromise system integrity.

****Security Impact:**** Compromise of system integrity and potential unauthorized access.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the AI component.

2. ****Threat ID:**** EA02

****Threat Name:**** Overprivileged External LLM Provider
****Affected Component/Data Flow:**** External LLM Provider, User Database
****Attack Scenario:**** The external LLM provider has excessive privileges that compromise system integrity.
****Security Impact:**** Compromise of user data and potential identity theft.
****Severity:**** Critical
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the External LLM Provider.

****Threats related to Session Security****

1. ****Threat ID:**** SS01

****Threat Name:**** Unauthorized Access to Session Storage
****Affected Component/Data Flow:**** P3, Session Storage
****Attack Scenario:**** An attacker gains unauthorized access to the session storage.
****Security Impact:**** Compromise of session security and potential access to sensitive data.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust security controls for the Session Storage.

2. ****Threat ID:**** SS02

****Threat Name:**** Session Tampering through AI Chatbot Component
****Affected Component/Data Flow:**** AI Chatbot Component, Session Storage
****Attack Scenario:**** The AI component is exploited to tamper with session data.
****Security Impact:**** Compromise of session security and potential access to sensitive data.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the AI component.

****Threats related to Authentication Abuse****

1. ****Threat ID:**** AA01

****Threat Name:**** Unauthorized Access to User Database through P2
****Affected Component/Data Flow:**** P2, User Database
****Attack Scenario:**** An attacker gains unauthorized access to the user database.
****Security Impact:**** Compromise of user data and potential identity theft.

****Severity:**** Critical
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the User Database.

2. ****Threat ID:**** AA02

****Threat Name:**** Unauthorized Access to Session Storage through P3
****Affected Component/Data Flow:**** P3, Session Storage
****Attack Scenario:**** An attacker gains unauthorized access to the session storage.
****Security Impact:**** Compromise of session security and potential access to sensitive data.
****Severity:**** High
****Likelihood:**** Medium
****Mitigation Strategy:**** Implement robust security controls for the Session Storage.

****Threats related to Data Leakage****

1. ****Threat ID:**** DL01

****Threat Name:**** Unauthorized Access to User Database through External LLM Provider
****Affected Component/Data Flow:**** External LLM Provider, User Database
****Attack Scenario:**** The external LLM provider gains unauthorized access to the user database.
****Security Impact:**** Compromise of user data and potential identity theft.
****Severity:**** Critical
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the External LLM Provider.

2. ****Threat ID:**** DL02

****Threat Name:**** Unauthorized Access to Session Storage through AI Chatbot Component
****Affected Component/Data Flow:**** AI Chatbot Component, Session Storage
****Attack Scenario:**** The AI component is exploited to gain unauthorized access to the session storage.
****Security Impact:**** Compromise of session security and potential access to sensitive data.
****Severity:**** High
****Likelihood:**** Low
****Mitigation Strategy:**** Implement robust security controls for the AI component.